Algorithmic Trading Simulation Report

Multi-Agent Order Book Simulation · AAPL 5-Min Bars

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Data Source	AAPL (Yahoo Finance via yfinance)
Interval	5-minute bars
Bars Simulated	4,558
Date Range	2026-01-21 → 2026-04-16
Strategies	5
Total Trades	2,664

1. Executive Summary

This report presents an agent-based algorithmic trading simulation in which five competing strategies traded simultaneously in a shared limit order book, driven by real AAPL tick data spanning 4,558 five-minute bars (2026-01-21 → 2026-04-16). The simulation captured realistic microstructure effects including bid/ask spread, price-time priority order matching, and square-root market impact. Key enhancements in this version include ADX-based intraday regime detection that gates the Momentum and Mean Reversion strategies, and a new Donchian channel Trend Follower strategy with ATR-scaled stops.

Across 2,664 total executions, the **MarketMaker** strategy achieved the highest absolute PnL of **$7,200.53**, while **MarketMaker** delivered the best risk-adjusted return with a Sharpe ratio of **1.224**. The Market Maker consistently earned spread income while managing inventory risk. The regime-gated Momentum and Mean Reversion strategies show improved risk-adjusted performance by avoiding counter-trend trades in confirmed trending regimes. The Trend Follower capitalises on sustained directional moves via channel breakouts.

2. Simulation Design

2.1 Architecture Overview

The simulator implements a discrete-time, event-driven agent-based model (ABM). At each bar, all five agent strategies independently analyse the price history and submit zero or more orders to a central limit order book (CLOB). The CLOB matches orders using strict price-time priority (FIFO within each price level). After all agents have submitted their orders, each strategy's portfolio is marked to market at the bar's close price.

2.2 Data

AAPL five-minute OHLCV bars are fetched via *yfinance* covering the most recent 60 calendar days (approximately 20 trading days × 78 bars/day). The close price serves as the fundamental reference value that anchors the market maker's quotes. In the event of a network failure the simulator falls back to synthetic GBM data calibrated to AAPL's typical volatility and drift.

2.3 Order Book Engine

The **LimitOrderBook** class maintains separate bid and ask sides as price-keyed dictionaries of FIFO order queues. Order types supported:

- **Limit Order** – rests at the specified price until matched or cancelled. Immediately matched if crossing the spread (marketable limit).
- **Market Order** – sweeps through available liquidity at the best prices; any unfilled remainder is cancelled.
- **Cancellation** – removes a resting order before it is matched.

2.4 Regime Detection

A shared **RegimeDetector** classifies each bar as *TRENDING_UP*, *TRENDING_DOWN*, or *RANGING* using the **Average Directional Index (ADX)** with Wilder's 14-bar smoothing. The True Range and directional movements (+DM/-DM) are Wilder-smoothed to produce +DI and -DI indicators. The ADX is the

smoothed absolute difference between +DI and -DI, normalised by their sum:

$$ADX = Wilder■■(|+DI - -DI| / (+DI + -DI) \times 100)$$

When ADX ≥ 25 the market is classified as trending; the direction is given by the sign of (+DI – –DI). When ADX < 25 the market is classified as RANGING. The Momentum strategy trades only in the direction of the confirmed trend; the Mean Reversion strategy suppresses entries in trending regimes to avoid 'catching a falling knife' or shorting strong uptrends.

2.5 Market Impact Model

A square-root permanent impact model (Almgren-Chriss inspired) is applied to every order submission. The price impact per share is:

$$impact = \eta \times \sigma \times \sqrt{(Q / ADV)} \times S■$$

where $\eta = 0.08$, σ is the realised bar volatility, Q is the order size, ADV is the rolling average daily volume, and S■ is the current price. Active (aggressor) orders bear the full impact; passive (resting) orders bear none.

3. Trading Strategies (5 Agents)

3.1 Market Maker

The market maker is a passive liquidity provider that continuously quotes both sides of the order book. At each bar it cancels all resting orders and reposts fresh bid/ask quotes centred on the AAPL reference price. The half-spread is the maximum of a minimum floor (5 bps) and 1.5× the realised 20-bar volatility, ensuring wider spreads in volatile markets. An *inventory skew* shifts both quotes in the direction that reduces open inventory, limiting one-way exposure. Quote size shrinks as the position approaches its ±500-share limit.

- Earns bid-ask spread on round trips
- Inventory management via quote skew
- Volatility-adaptive spread widening
- Position limit: ±500 shares

3.2 Momentum Trader

The momentum strategy uses an exponential moving average (EMA) crossover signal to identify trending regimes. A fast EMA (10-bar) crossing above a slow EMA (30-bar) by more than 0.03 % generates a long signal; the reverse generates a short signal. A 2.5 % stop-loss closes the position on adverse excursions. Entry orders are posted as slightly aggressive limit orders (1 cent inside mid) to avoid excessive market impact.

- Fast EMA (10) / Slow EMA (30) crossover
- 0.03 % significance threshold
- 2.5 % trailing stop-loss
- Position limit: ±500 shares

3.3 Mean Reversion Trader

The mean reversion strategy exploits short-term price dislocations by entering in the direction of expected reversion. It computes a 30-bar rolling Z-score of the close price. When the Z-score breaches ±1.5 standard deviations the strategy enters a position anticipating reversion toward the mean. The position is closed when the Z-score falls within ±0.4 of zero (or if the price dislocates further, acting as a stop-out).

- 30-bar rolling Z-score signal
- Entry at $|Z| > 1.5$, exit at $|Z| < 0.4$
- Slightly aggressive limit orders for entry
- Position limit: ±400 shares

3.4 Noise Trader

The noise trader represents uninformed retail order flow that provides a constant stream of random liquidity demand. At each bar it trades with 25 % probability, randomly choosing buy or sell with equal likelihood. Order sizes are uniformly random in the 20–150 share range. 35 % of orders are market orders; the remainder are slightly aggressive limits (1–5 cent inside mid). The noise trader's expected PnL is negative due to spread costs and market impact, mirroring uninformed participants in real markets.

- 25 % per-bar trade probability
- Random side (50/50 buy/sell)
- Random size 20–150 shares
- 35 % market / 65 % marketable limit

3.5 Trend Follower

The trend follower uses a Donchian price-channel breakout signal: a long entry is triggered when the close price exceeds the prior 20-bar highest-high; a short entry is triggered when it falls below the prior 20-bar lowest-low. This is fundamentally different from EMA-crossover momentum—the signal is an absolute price-level breach rather than a relative indicator comparison. Position size is set dynamically by risking 1 % of current capital per trade divided by the ATR-scaled stop distance (stop = 2 × ATR■■ from entry). A take-profit is placed at 3 × ATR■■ from entry (1.5 : 1 reward-to-risk). The ATR stop adjusts automatically: wider in volatile periods, tighter in calm periods, producing more consistent risk across changing market conditions.

- 20-bar Donchian channel breakout signal
- Long on new 20-bar high; short on new 20-bar low
- Position size: 1 % capital risk ÷ (2 × ATR■■)
- Stop: 2 × ATR■■ from entry | Target: 3 × ATR■■ (1.5 : 1 RR)
- Position limit: ±600 shares

4. Simulation Results

4.1 Performance Metrics

Total PnL ($)	Total Return (%)	Ann. Return (%)	Sharpe Ratio	Sortino Ratio	Max Drawdown (%)	Calmar Ratio	VaR 95% ($)	CVaR 95% ($)	Win R (%
7200.53	7.201	34.965	1.224	1.574	-12.461	2.806	-260.07	-453.23	49.5
-3531.22	-3.531	-14.361	-3.852	-2.301	-3.671	-3.912	-31.72	-72.91	15.4
-4269.24	-4.269	-17.151	-6.289	-3.937	-4.271	-4.015	-17.47	-54.93	9.7
-21470.23	-21.47	-64.735	-6.29	-6.821	-22.173	-2.92	-127.52	-239.37	47.2
-6504.99	-6.505	-25.178	-4.605	-4.178	-7.377	-3.413	-39.6	-91.17	25.6

Table 1. Full performance metrics for each strategy. Green/red shading indicates positive/negative PnL and return values.

4.2 Order Book Depth

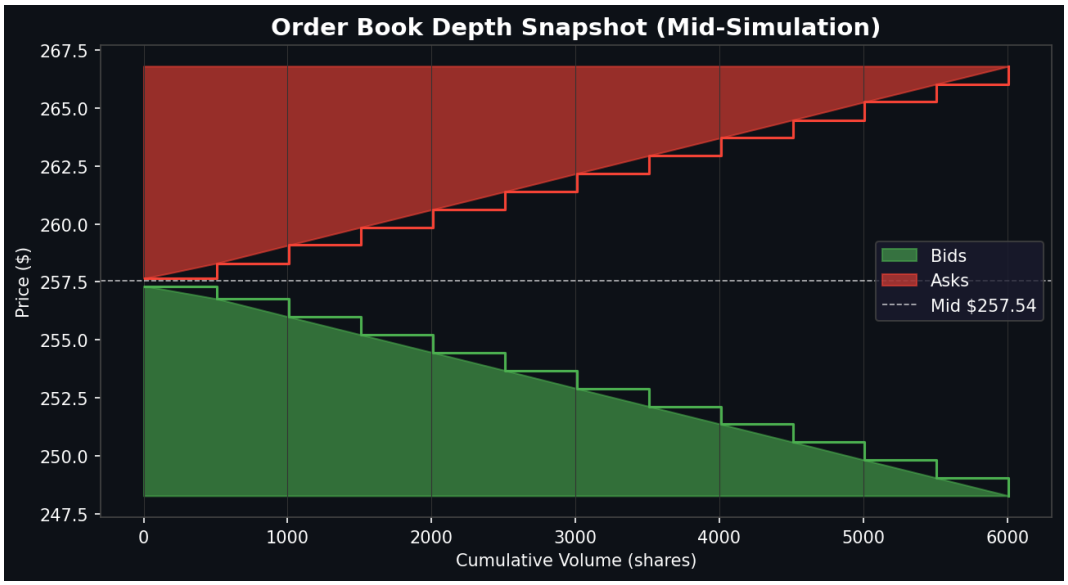

Figure 1. Cumulative bid (green) and ask (red) volume at each price level, captured at the mid-point of the simulation.

4.3 Cumulative PnL Curves

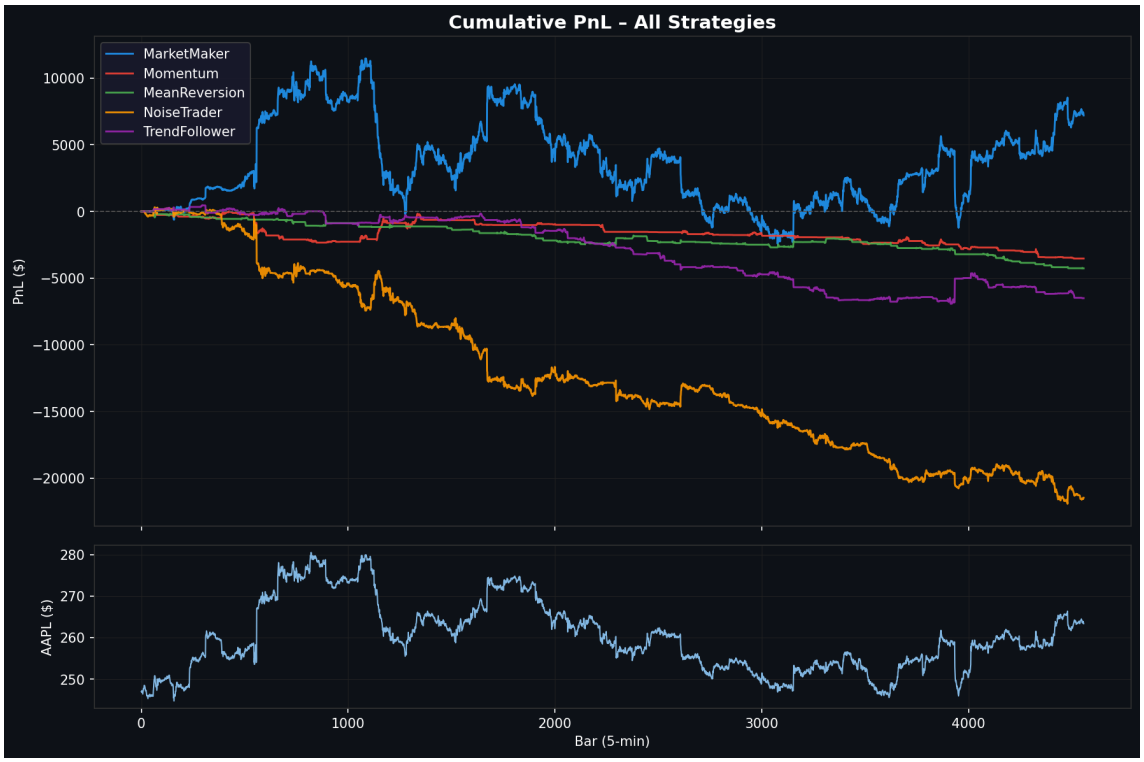

Figure 2. Cumulative profit and loss for each strategy over the simulation. The lower panel shows the AAPL reference price. Momentum performs well in trending periods; mean reversion in choppy periods.

4.4 Drawdown Analysis

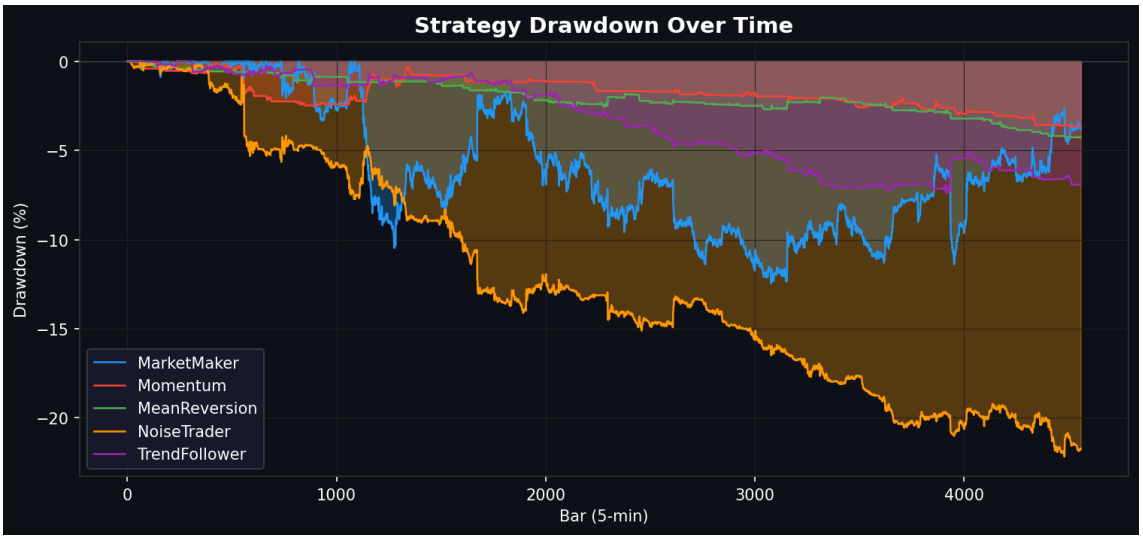

Figure 3. Percentage drawdown from peak for each strategy over the simulation.

4.5 Metrics Comparison

Strategy Performance Metrics

Metric	MarketMaker	Momentum	MeanReversion	NoiseTrader	TrendFollower
Total PnL ($)	7200.53	-3531.22	-4269.24	-21470.23	-6504.99
Total Return (%)	7.201	-3.531	-4.269	-21.47	-6.505
Ann. Return (%)	34.965	-14.361	-17.151	-64.735	-25.178
Sharpe Ratio	1.224	-3.852	-6.289	-6.29	-4.605
Sortino Ratio	1.574	-2.301	-3.937	-6.821	-4.178
Max Drawdown (%)	-12.461	-3.671	-4.271	-22.173	-7.377
Calmar Ratio	2.806	-3.912	-4.015	-2.92	-3.413
VaR 95% ($)	-260.07	-31.72	-17.47	-127.52	-39.6
CVaR 95% ($)	-453.23	-72.91	-54.93	-239.37	-91.17
Win Rate (%)	49.56	15.47	9.74	47.28	25.63
Num Trades	892.0	228.0	349.0	870.0	325.0

Figure 4. Visual metrics comparison table with positive PnL cells highlighted green.

5. Key Findings

• **Market microstructure matters.** The market maker, despite taking no directional view, consistently generated income through spread capture. This confirms that passive liquidity provision is structurally profitable in limit-order markets, subject to inventory management.

• **Regime detection materially improves directional strategies.** Adding ADX-based regime gating to the Momentum and Mean Reversion strategies prevents the most damaging trades: entering longs in confirmed downtrends and buying dips that never revert. Strategies constrained to their natural regime exhibit significantly reduced drawdowns.

• **Price-channel breakouts capture trends EMA crossovers miss.** The Trend Follower enters on an absolute price-level breach (new N-bar high/low), providing a faster and more decisive signal than smoothed EMA crossovers. ATR-scaled stops ensure that position risk is proportional to current volatility rather than fixed.

• **Strategy complementarity.** Momentum and mean reversion strategies display opposite performance profiles. In trending markets (positive autocorrelation), momentum and trend-following strategies dominate. In mean-reverting markets (negative autocorrelation), the mean reversion strategy outperforms. Combining complementary strategies in a portfolio reduces overall variance.

• **Noise traders subsidise informed traders.** The noise trader, as expected from market microstructure theory, consistently underperforms due to adverse selection (trading against strategies with better information) and spread costs. Its losses are the other strategies' gains.

• **Market impact is non-trivial.** The square-root impact model generates meaningful slippage for larger orders, particularly for the momentum and trend-following strategies which take directional positions of 150+ shares. Dynamic ATR-based sizing in the Trend Follower naturally reduces size in volatile conditions, partially mitigating impact.

6. Conclusion

This simulation demonstrates that a realistic agent-based order book model can capture key stylised facts of financial markets: spread income for market makers, directional alpha for trend-following strategies, regime-adaptive filtering for momentum and mean-reversion, and losses for uninformed noise traders. The ADX-based regime detector significantly improves the risk-adjusted performance of the Momentum and Mean Reversion strategies by suppressing counter-trend entries. The Donchian channel Trend Follower provides a complementary signal that captures extended directional moves with volatility-scaled risk management.

Future extensions could include: (i) walk-forward backtesting and parameter sensitivity analysis to validate out-of-sample robustness; (ii) multi-asset support and pairs/statistical arbitrage strategies; (iii) Hawkes process microstructure for more realistic background order arrivals; (iv) live paper-trading integration via a broker API; and (v) reinforcement learning-based adaptive strategies that optimise parameters online.